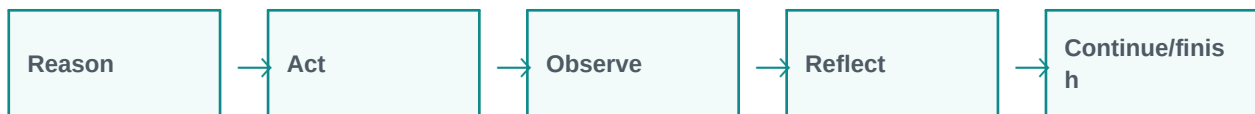


Planning, ReAct, and Reasoning Loops

Reasoning loops let agents work step by step.

Simple explanation: Planning is breaking a big task into smaller steps. ReAct is a common agent pattern where the model reasons, acts using a tool, observes the result, and continues.

Learning style: this handout starts with a very simple explanation and gradually increases the depth. It is designed for classroom training, participant reading, and quick revision.



What you will learn

- Meaning in simple words
- Key components and architecture
- Practical examples and use cases
- Where it fits in GenAI and agentic AI systems
- Benefits, challenges, and implementation points



1. Beginner-Friendly Explanation

Planning is breaking a big task into smaller steps. ReAct is a common agent pattern where the model reasons, acts using a tool, observes the result, and continues.

Think of it like this:

A reasoning loop is like solving a problem step by step: think, do, check, improve, and continue.

Simple real-time examples

- Debugging workflow
- Research workflow
- Data analysis workflow
- Travel planning
- Complex document creation

2. Gradually Deeper Explanation

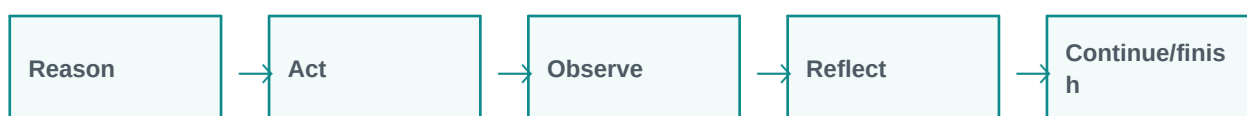
Planning, ReAct, and Reasoning Loops becomes more powerful when we understand its components, data flow, limitations, and business fit. In real projects, the concept is rarely used alone. It usually works with prompts, models, tools, documents, APIs, evaluation, monitoring, and human review.

Important concepts

Concept	Meaning
ReAct loop	ReAct loop is an important part of understanding and applying Planning, ReAct, and Reasoning Loops in practical GenAI systems.
Plan-and-execute	Plan-and-execute is an important part of understanding and applying Planning, ReAct, and Reasoning Loops in practical GenAI systems.
Reflection	Reflection is an important part of understanding and applying Planning, ReAct, and Reasoning Loops in practical GenAI systems.
Self-checking	Self-checking is an important part of understanding and applying Planning, ReAct, and Reasoning Loops in practical GenAI systems.
Task decomposition	Task decomposition is an important part of understanding and applying Planning, ReAct, and Reasoning Loops in practical GenAI systems.

3. Architecture / Flow

A typical architecture can be understood as a simple flow. The exact design changes based on the project, but the pattern below is a useful starting point.



Architecture explanation

Step 1: Reason - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 2: Act - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 3: Observe - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 4: Reflect - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 5: Continue/finish - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Common implementation layers

- User interface or API layer
- Prompt and instruction layer
- Model layer
- Tool/retrieval/data layer
- Validation, safety, and logging layer
- Human review or feedback layer

4. Detailed Examples

Below are example scenarios showing how this topic appears in real GenAI projects.

Scenario	How it works	Expected output
Debugging workflow	The system receives the request, applies Planning, ReAct, and Reasoning Loops, and processes the task step by step.	A useful, structured, reviewable result.
Research workflow	The system receives the request, applies Planning, ReAct, and Reasoning Loops, and processes the task step by step.	A useful, structured, reviewable result.
Data analysis workflow	The system receives the request, applies Planning, ReAct, and Reasoning Loops, and processes the task step by step.	A useful, structured, reviewable result.



Scenario	How it works	Expected output
Travel planning	The system receives the request, applies Planning, ReAct, and Reasoning Loops, and processes the task step by step.	A useful, structured, reviewable result.
Complex document creation	The system receives the request, applies Planning, ReAct, and Reasoning Loops, and processes the task step by step.	A useful, structured, reviewable result.

5. Benefits and Challenges

Benefits	Challenges / Risks
Improves productivity and reduces repetitive manual effort.	Can produce incorrect or unsupported answers if not validated.
Helps users interact with complex systems in natural language.	Needs careful design of prompts, data access, permissions, and monitoring.
Can support training, support, coding, analysis, and document workflows.	May have cost, latency, privacy, and governance concerns in production.

6. Best Practices

- Start with a simple prototype before building a complex system.
- Use clear prompts and structured output formats.
- Add validation and human review for important decisions.
- Measure quality with realistic test cases, not only demo examples.
- Monitor cost, latency, accuracy, safety, and user feedback.
- Document assumptions, limitations, and escalation paths.

7. Key Takeaways

Planning, ReAct, and Reasoning Loops is an important concept in modern LLM and GenAI systems. The simple idea is easy to understand, but production implementation requires thoughtful architecture, data quality, evaluation, safety, and monitoring.

Quick revision

- Simple meaning: Planning is breaking a big task into smaller steps. ReAct is a common agent pattern where the model reasons, acts using a tool, observes the result, and continues.
- Architecture: input, processing, tools/data, validation, output.
- Business value: saves time, improves knowledge access, and supports automation.
- Risk control: use grounding, evaluation, guardrails, monitoring, and human approval.